UEC512: LINEAR INTEGRATED CIRCUITS AND APPLICATIONS

L	T	P	Cr
2	0	2	3.0

Course Objective: Course Objective: To enhance comprehension capabilities of students through understanding of operational amplifiers, frequency response, various applications of operational amplifiers, active filters, oscillators, analog to digital and digital to analog converters and few special function integrated circuits.

Introduction to Differential Amplifiers: Differential Amplifier, configurations of differential amplifier, Analysis of single input balanced output, single input unbalanced output, dual input balanced output and dual input unbalanced output differential amplifiers

Operational amplifier: various characteristics of op-amp, CMRR, PSRR, Internal structure of Op-amp, Ideal Op-amp.Inverting and Non-Inverting Configuration, Ideal Open-Loop and CLO ssed-Loop Operation of Op-Amp, Feedback Configurations: Voltage-Series Feedback Amplifier, Voltage-Shunt Feedback Amplifier, Differential Amplifiers with One & Two Op-Amps

Frequency Response of an Op-Amp: Introduction to Frequency Response, Compensating Networks, Frequency Response of Internally Compensated Op-Amp, Frequency response of Non-compensated Op-Amp, CLO ssed-Loop Frequency Response.

General Applications: DC & AC Amplifiers, Peaking Amplifier, Summing, Scaling and Averaging amplifier, Instrumentation Amplifier, The Integrator, The Differentiator, Log and Antilog Amplifier, Comparator, Zero Crossing Detector, Schmitt Trigger, Sample and Hold Circuit, Clippers and Clampers etc.

Active Filters and Oscillators: Butterworth Filters, Band-Pass Filters, Band Reject Filters, All Pass Filters, Phase Shift Oscillator, Wien Bridge Oscillator, Voltage-Controlled Oscillator (VCO), Square Wave Generator.

Specialized IC Applications: Introduction, The 555 Timer, Monostable and Astable Multivibrator using IC 555, Phase-Locked Loop (PLL), Voltage Regulators.

Familiarization with Standards: IEEE 165-1977.

Laboratory Work: Inverting and Non-Inverting Characteristics of an Op-Amp, Measurement of Op-amp parameters, Op-amp as integrator & differentiator, comparator, Schmitt trigger, Converter (ADC, DAC), square wave generator, Sawtooth waveform generator, precision half wave and full wave rectifiers, log-antilog amplifier, 555 as an astable, monostable and bi-stable multivibrators, active filters.

Course Learning Objectives (CLO)

The students will be able to:

1. Analyse differential amplifier and its configurations.
2. Explain various op-amp parameters and configurations.
3. Analyse frequency response of op-amps
4. Design various applications using op-amps
5. Implement circuits using Linear IC (555)

Text Books

1. Ramakant A. Gayakwad, „OP-AMP and Linear IC‟s‟, Prentice Hal, 1999.
2. Sergio Franco, „Design with operational amplifiers and analog integrated circuits‟, McGraw-Hill, 2002.

Reference Books

1. D. Roy Choudhry, Shail Jain, “Linear Integrated Circuits”, New Age International Pvt. Ltd., 2000.
2. J. Michael Jacob, „Applications and Design with Analog Integrated Circuits‟, Prentice Hall of India, 2002.